

Feature 0: Fundamentals

Application Infrastructure & Architecture Fundamentals

User Stories

- **As a Staff Member**, I want to see assigned tasks in real-time, receive audible notifications for new assignments, and update task statuses effortlessly.
- **As an Admin**, I want to assign tasks, monitor real-time operational activity, and push urgent instructions via notifications.
- **As Management**, I want to extract performance data and view visual dashboards to track operational progress and productivity.
- **As the IT/Infrastructure Team**, I want the application to pass network security testing and be compatible with MDM white-listing for corporate mobile device deployment.

Functional Requirements

1. **Multi-Platform Support:** Native development for both Android and iOS across all current versions.
2. **Role-Based Interfaces:**
 - **Staff Interface:** Minimalist UI focused on task lists and completion actions.
 - **Admin Interface:** Activity overview, assignment management, and messaging tools.
 - **Management Interface:** Reporting tools and visual analytics dashboards.
3. **Communication Engine:** Integrated notification system with distinct audible alerts for new task assignments or critical updates.
4. **Security & Compliance:**
 - Adherence to industry-standard network security protocols.
 - Compatibility with enterprise MDM solutions for application white-listing on corporate devices.
5. **Languages:** In French and English for the staff and admin. Management and data should be only in English.
6. **Admin:** should be able to create and remove the accesses and assign different access to different member. Should be able to design the shift hours and days of working per staff.

Process Flow Logic

1. **Authentication & Compliance:**
 - User initiates login; system verifies MDM compliance (device white-listing) and valid credentials.
2. **Interface Routing:**
 - The system redirects users based on role (Staff, Admin, or Management).

3. Task & Communication Loop:

- Staff interface polls/receives pushes for new assignments.
- Notification triggers audio alert upon reception.
- Staff interacts with UI to update task progress (start/complete/number of what processes where applicable).

4. Data Synchronization & Reporting:

- Real-time activity logs are pushed to the centralized database.

Management dashboards aggregate this data to visualize advancement and efficiency metrics.

Feature 1: Attendance

Feature 1: Shift Attendance (Clock-In/Clock-Out) - MVP

This feature replaces manual chat notifications with a formal attendance tracking system, ensuring accurate time logging and simplifying payroll reporting.

User Stories

- **As a Staff Member**, I want to clock in and out within the application so that my working hours are automatically recorded without needing to post in a chat group.
- **As an Admin**, I want to view a real-time log of who is currently working and generate reports on total hours worked by each staff member.

Functional Requirements

1. **Dashboard Integration:** A prominent "Start Shift" or "End Shift" button must be accessible immediately upon user login/log out.
2. **State Management:**
 - The button label must toggle: "Start Shift" (when inactive) to "End Shift" (when active).
 - The system must capture the precise timestamp (UTC and Local Time) upon the user's action documented in a Google Sheet.
3. **Authentication/Validation:**
 - The action must be tied to the unique user ID.
 - Prevent duplicate logs (e.g., cannot clock in twice without clocking out).
4. **Logging Mechanism:** All clock-in/clock-out events must be stored in a secured database table with: **User ID**, **Action Type** (IN/OUT), **Timestamp**, and **Device IP** (optional for audit).
5. **Notifications:** (Optional) If the user attempts to log out without completing a specific daily requirement, a warning message should appear.

Process Flow Logic

You can visualize this flow as a standard "State Machine" diagram:

1. **User Login:** User authenticates into the application.
2. **Check Current Status:**
 - System queries the database: *Is there an active session for this User ID?*
 - **If YES:** Show "End Shift" button.
 - **If NO:** Show "Start Shift" button.
3. **Action Initiation:**
 - **Case A (Start Shift):** User clicks "Start Shift".
 - System writes [Timestamp] + [Start_Event] to database.
 - System UI updates: Button changes to "End Shift".
 - **Case B (End Shift):** User clicks "End Shift".
 - System writes [Timestamp] + [End_Event] to database.
 - System UI updates: Button changes to "Start Shift".
4. **Completion:** System displays a confirmation toast message (e.g., "Shift started at 09:00 AM").
5. The User receives a notification when 7 hours of work is completed to be able to log out of the system. Working hours are 7:30 to 15:30 or 9:00 to 17:00. If the end of 7 hours is before end hours, the user will receive the notification only at the defined ending hour. But if the end of 7 hours is after cut-off time, the user receives a message mentioning that he needs to work until xx:xx hour to complete the 7-hour shift.

Extra features for Attendance - Future developments

This update integrates mandatory location verification (geofencing) and automated break scheduling into the existing attendance framework. These additions ensure operational compliance and accurate time tracking for payroll.

1. Location Verification (Geofencing)

- **Coordinate Registration:** The Admin must define the Office Location (Latitude/Longitude) and an allowed "Accuracy Radius" (e.g., 50 meters).
- **Permission Check:** The application must request location permissions on login. If denied, the system prohibits "Start Shift" or "End Shift" actions.
- **Validation Logic:** Upon clicking "Start Shift," the system calculates the distance between the user's device location and the registered Office Location.
- **Restriction:** If the distance exceeds the Allowed Radius, the action is blocked, and an error message is displayed: *"You must be at the office location to clock in."*

2. Automated Break Management

- **Schedule Definition:** Admin defines daily break windows and duration (e.g., 10:30 AM, 1:00 PM, 3:30 PM).
- **Notification Engine:** The system triggers a push notification to the user 5 minutes prior to the scheduled break time.
- **Break Types:**
 - **Paid Breaks (10m):** Logged as "Work Active" but flagged as "On Break" to maintain continuity in shift status.
 - **Unpaid Lunch (60m):** Triggers an automatic "Pause Shift" event (clock stops) and requires an "End Break" action to resume the shift. The user receives notifications to start/end lunch break 10 minutes before the cut-off time.

Updated Process Flow Logic

Phase 1: Attendance Verification

1. **Initiation:** User clicks "Start Shift."
2. **Location Check:** System requests real-time coordinates.
3. **Validation:**
 - **Match:** Proceed to log the shift start.
 - **Mismatch:** Abort action, prompt user to move to the office, or provide an "Override" field for Admin approval.
4. **Entry:** Write [Timestamp] + [User_ID] + [Coordinates] + [Status:

Started] to the database.

Phase 2: Break Management Loop

1. **Monitor:** A background service constantly compares the current time against the Admin-defined break schedule.
2. **Notify:** At [Break_Time - 5 mins], send alert: "Reminder: Your break starts in 5 minutes."
3. **Execution:**
 - **For Paid Breaks:** System logs [Status: On Paid Break] at [Start_Time] and auto-resumes to [Status: Working] after 10 minutes.
 - **For Unpaid Breaks:** System logs [Status: On Lunch] at [Start_Time] and pauses the time tracking calculation until the user manually triggers "Resume Work."

Phase 3: End of Shift

1. **Validation:** Perform Location Check again (optional, depending on security requirements).
2. **Termination:** Log [Timestamp] + [Status: Ended].
3. **Calculation:** Total Working Time = (Total Time on Premises) - (Total Unpaid Break Time).

To track the efficiency of your warehouse operations, we need to introduce a "Process Completion Timestamp" to the Reception module. This will allow the system to calculate the duration between the initial arrival and the final storage confirmation.

Feature 2: Reception

Feature 2: Reception - Incoming Flow

This feature replaces manual chat messages with a formal reception tracking system, ensuring accurate time logging of reception and put-aways.

User Stories

- **As a Staff Member**, I want to inform admin and management of the arrival of trucks and the nature of what was received and get the instruction of how and where to put received parcels/pallets.
- **As an Admin**, I want to view a real-time log of what is received and give instructions and document how long one reception of each kind would take to reach to putaway status.

Functional Requirements

1. Dual-Timestamp Logging

- **Arrival Timestamp (Start)**: The initial "Record Arrival Time" button remains the start point.
- **Put-Away Timestamp (End)**: The system must automatically capture a second timestamp the exact moment the user clicks the "Confirm Put Away" checkbox or "Submit" button.

All recorded in a Google sheet.

2. Analytics Calculation

- **Formula**: The system will calculate:
$$\text{Processing Time} = [\text{Put-Away Timestamp}] - [\text{Arrival Timestamp}]$$
- **Reporting**: The Admin dashboard should display this duration for each entry, allowing for an "Average Processing Time" report per reception category.

Process Flow Logic

1. **Step 1: Arrival (Start of Timer)**
 - User clicks "New Delivery."
 - System records `[Arrival_Timestamp]` in the `Reception_Log` database with `Status: Arrived`.
2. **Step 2: Reception & Data Entry**
 - User performs the specific category flow (selecting type, entering parcel count or material description, adding remarks).
 - Categories:

- i. **Return Parcels:** data to be collected for this category: number of parcels received + the transporter company
- ii. **Packaging Stock:** data to be collected for this category: number of parcels/pallets received + type of packaging stock received
- iii. **Sellers Stock:** data to be collected for this category: number of pallets received + see Feature 3 for details
- iv. **Equipment and Other Parcels:** data to be collected for this category: number of parcels received + type of items received in message box + photo of what received

3. Step 3: Get Instructions from the Admin

- Admin investigates the reception, give a go ahead for the received parcels / pallets
- The user receives a notification with instructions of what to do with the received parcels/pallets.

4. Step 4: Completion (Stop of Timer)

- User checks the box: "Confirm items are put away."
- **Crucial Action:** The system triggers an automatic `[Put_Away_Timestamp]` (Current System Time).
- The system calculates the duration between the two timestamps.
- The database record is updated: `Status: Completed` + `Processing_Duration: [Value]`.

5. Step 5: Audit (Optional)

- If the processing time exceeds a pre-defined threshold (e.g., 2 hours), the system can flag the entry in the Admin dashboard for review.

Feature 3: Seller Stock

Feature 3: Seller Stock Reception - Incoming Flow

This feature handles the intake of high-volume stock, enforcing weight checks and quality control to ensure data accuracy and operational safety before the stock moves to the pipeline for inbound.

The **Seller Stock Reception** module requires more robust logic because it involves physical constraints (weight) and quality control branching (damaged vs. good).

User Stories

- **As Warehouse Staff**, I want to log pallet details and condition so that I can quickly process safe pallets and flag damaged pallets for administrative review.
- **As an Admin**, I want to see photos of shipping labels, pallet conditions, and damage evidence so that I can make informed decisions on how to handle non-compliant stock. And also to be able to automatically generate the damage reports for the supply team as well as the seller.
- **As an Operations Manager**, I want items to automatically move to the "Put-Away Pipeline" once they are processed so that they can be cataloged online efficiently.

Functional Requirements

1. **Mandatory Data Entry:**
 - **Label Capture:** Required photo of the shipping label.
 - **Seller Name:** Mandatory entry of shipper (seller name) per pallet for indexing.
 - **Pallet ID:** Mandatory entry of one box number per pallet for indexing.
 - **Dual Data Pairing:** The system must generate a unique "Pallet Index" to link the physical pallet to the digital record for admin searchability.
 - **Weight Entry:** Manual input for pallet weight.
2. **Conditional Logic (Quality Gate):**
 - **Threshold:** If Weight > 700kg, trigger an "Overweight Warning" (Flag for manual review).
 - **Condition Status:**
 - **Good:** Proceed to "Ready for Put-Away."
 - **Damaged:** Halt process, force entry of "Damage Remarks," and require mandatory photo evidence of the damage.
3. **Pipeline Transition:**
 - When the staff completes documentation for all pallets, the status must update from **In_Reception** to **Ready_For_Pipeline**, moving the records to the **Stock Put-Away Pipeline** tab.

Process Flow Logic

1. Initial Scan:

- User selects **Stock Reception - Category : Seller Stock**
- Camera activates: User captures **Shipping_Label_Photo**.
- User enters **Box_Number**.
- User enters **Seller_Name**.

2. Measurement & Inspection:

- User inputs **Pallet_Weight**.
- User selects **Condition** (Dropdown: Good / Damaged).

3. Branching Decision:

- **Branch A (Good Condition & < 700kg):**
 - System tags record: **Status: Ready_For_Put_Away**.
 - Notification sent to Admin.
 - Record is visible in the **Stock Put-Away Pipeline** tab.
 - The user receives a notification from the admin to instruct where the pallet goes in the warehouse.
- **Branch B (Damaged or > 700kg):**
 - System flags record: **Status: Pending_Admin_Review**.
 - Mandatory: User inputs **Damage_Remarks** and captures **Damage_Evidence_Photos**.
 - Notification sent to Admin.
 - The user receives a notification from the admin to instruct where the pallet goes in the warehouse.

4. Batch Completion:

- Staff marks the batch of pallets as **Documentation_Complete**.
- System transfers all **Ready_For_Put_Away** items to the **Pipeline** tab.

Feature 4: Seller Stock Location Assig...

Feature 4: Seller Stock Put-Away & Location Assignment

This feature optimizes warehouse efficiency by managing the assignment of stock placement tasks, tracking labor time, and providing a mechanism for exception handling during the put-away process.

User Stories

- **As an Admin**, I want to assign specific stock placement tasks to staff based on warehouse capacity and stock status so that inventory is organized efficiently.
- **As Warehouse Staff**, I want to receive clear instructions on where to place pallets so that I can perform my tasks without ambiguity.
- **As an Operations Manager**, I want to log start and end timestamps for each pallet placement so that I can analyze productivity and identify bottlenecks.

Functional Requirements

1. **Task Assignment Dashboard:**
 - Admin interface to assign specific pallet locations (racks/dedicated areas) to designated staff.
2. **Notification & Instructions:**
 - Push notification system for staff to receive real-time task details, including pallet ID and target location.
3. **Time Tracking:**
 - System-triggered timestamps: "Start" (when the user begins processing) and "Complete" (when the user finishes the specific pallet).
4. **Exception Handling (Issue Reporting):**
 - A "Report Issue" interface for staff to flag blockers (e.g., heavy items, incorrect location data) for Admin intervention.
5. **Performance Analytics:**
 - Automated logging of task duration, segmented by stock status and flags, to support management review.

Process Flow Logic

1. **Assignment:**
 - Admin reviews the "Ready For Put-Away" list and assigns tasks to specific staff members.
2. **Notification & Start:**
 - Staff receives a notification with destination details.

- Staff initiates the task, triggering the "Start" timestamp.
- 3. **Execution & Branching:**
 - **Branch A (Successful Completion):**
 - Staff completes the placement.
 - Staff clicks "Task Completed," recording the "End" timestamp and fetching the next task.
 - **Branch B (Exception/Problem):**
 - Staff selects "Report Issue."
 - The user provides context for the block (e.g., "Too many heavy boxes").
 - Notification sent to Admin for further instructions.
- 4. **Data Aggregation:**
 - System updates records with total time elapsed, enabling admin reporting on throughput efficiency.

Feature 5: Order Preparation

Feature 5: Order Preparation (Picking & Packing Optimization)

This feature optimizes the end-to-end order processing flow by balancing staffing resources for picking and packing. It leverages throughput data to calculate optimal labor allocation, dynamically staggering work starts to ensure a continuous, efficient pipeline while capturing detailed performance metrics.

User Stories

- **As an Admin**, I want to input the volume of parts to be processed so that the system can automatically calculate the optimal number of pickers and packers required for the day.
- **As an Operations Manager**, I want to track start and end timestamps for every task so that I can evaluate individual staff efficiency and update our hourly performance averages based on actual data.
- **As a Warehouse Staff Member**, I want to receive task assignments that reflect the actual workload so that I can work consistently without unnecessary downtime.

Functional Requirements

1. **Labor Optimization Calculator:**
 - Admin interface to input total parts to be picked and packed.
 - System calculates staff requirements based on baseline throughput (Pickers: 25 parts/hr; Packers: 20 parcels/hr).
2. **Dynamic Workflow Staggering:**
 - The system intelligently determines the "packing start time" relative to the "picking start time" to prevent packer idle time, ensuring a smooth handoff between picking and packing stations.
3. **Task Time Tracking:**
 - Automated logging of "Start" and "End" timestamps for every individual pick or pack task performed by staff.
4. **Performance Analytics & Updates:**
 - System aggregates task duration data to refine the average throughput metrics (parts/hr) and identify high-performing staff combinations.

Process Flow Logic

1. **Planning:**
 - Admin inputs the total volume of parts.
 - System calculates required personnel and generates a shift schedule.

2. Initialization:

- System generates start times for Pickers. Pickers confirm they started the task assigned.
- System calculates the lead time (delay) required for the first batch of items to be picked and send a notification to the pickers that they should pick xx parts until hh minutes.
- System sets the Packing start time accordingly to avoid idle labor and when enough parts are picked, packer receives notification to start packing and confirms that they started.

3. Execution & Data Capture:

- Staff executes picking/packing tasks.
- Each scan or task completion records a timestamp.

4. Continuous Improvement:

- Upon task completion, the system updates actual throughput records.
- Management dashboard visualizes efficiency, helping to determine the most effective staff combinations for future shifts.

Returns and Errors Documentation

documentation of returns with wrong packaging shipping etc

documentation of errors

Onglet 7

outgoing flow
return process

Onglet 9

Reporting and visualisation